Chapter 7 Student Notes and Labs

Emotion and the Body: How Internal States Tell the Brain What Matters

Student Notes

What this chapter is for

Chapter 6 built a brain that infers what is out there. Chapter 7 asks the next question every animal must answer: does it matter, and how much? The answer cannot come from the world. It comes from the body the brain is built to keep alive, and emotion is the brain's running estimate of the state of that body and of what the current situation means for it.

If you come from software or machine learning, the chapter is designed to correct three instincts:
1) treating emotion as noise that interferes with reasoning, rather than the part of the model that says what matters,
2) treating value or reward as a fixed property of an outcome, rather than something that depends on the state of the agent, and
3) treating regulation as reaction to error, rather than prediction of need.

Your goal is to be able to explain, in plain language, how homeostasis and allostasis differ, how signals from the body reach the brain and what the insula does with them, what the James-Lange theory claimed and what has happened to it, how the basic emotion, constructionist, and survival-circuit views differ, why the value of food depends on hunger, how the amygdala learns threat and how extinction suppresses it, how the two arms of the stress response work and why their effects follow an inverted U, how arousal decides what is remembered, and what the Iowa gambling task does and does not show.

What perception leaves out

A model of the world is not yet a reason to act. Value is relative to the state of the body and its prospects, so the brain needs a second model alongside its model of the world: a model of the body itself. Chapter 1 described the brain as a controller keeping internal variables within bounds; this chapter takes that literally.
Thesis (one that most competing theories of emotion can accept): emotions are, at their core, an estimate of the state of the body and what the situation means for it, and they do three jobs at once: prepare the body, rank the options for action, and tell the learning machinery of Chapters 3 and 5 what to keep.

Keeping the body in bounds

- Claude Bernard (1850s and 1860s): cells live in the *milieu intérieur*, which must be held constant. Walter Cannon named its maintenance homeostasis in the 1920s.
- Narrow ranges: core temperature within about a degree of 37 °C; arterial pH 7.35 to 7.45; fasting glucose about 4 to 6 mM (the brain runs mostly on glucose in the fed state). Each is defended by negative feedback: sensor, set point, effector.

- The hypothalamus is the hub. It reads temperature, osmolarity, and glucose, and corrects through three channels: the autonomic nervous system, the pituitary, and behavior (hunger, thirst, restlessness).
- Behavior is where regulation becomes emotion. Cannon and Britton (1925): cats without cortex show "sham rage"; Bard (1928): it is abolished when the posterior hypothalamus is removed. Lin and colleagues (2011): stimulating a small set of ventromedial hypothalamic neurons makes male mice attack whatever is in front of them, a rubber glove included.

Regulation by prediction

- Much regulation is feedback: on standing, the baroreflex corrects the fall in blood pressure within a heartbeat or two. But the body also acts ahead of need. Krogh and Lindhard (1913): heart rate and breathing rise within a second or two of starting exercise, before any chemical signal of need.
- Allostasis (Sterling and Eyer 1988): stability through change; anticipate need and adjust in advance (Figure 7.1).
- Cephalic phase insulin release: sight, smell, and taste of food trigger insulin before glucose is absorbed. Woods (1991): eating would be dangerously destabilizing without it.
- Chen and colleagues (2015): hunger-promoting AgRP neurons in the arcuate nucleus fall silent within seconds of a hungry mouse seeing or smelling food, before it eats. The signal tracks the predicted state.
- Siegel and colleagues (1982): heroin overdose test. Drug-naive rats 96 percent died; experienced rats in the familiar room 32 percent; experienced rats in a new room 64 percent. Part of tolerance is learned anticipation tied to cues.
- Ramsay and Woods (2014) argue that allostasis is homeostasis with good prediction, not a new principle. The disagreement is about vocabulary; both sides accept that regulation is largely predictive and learned, the same strategy Chapter 6 described for perception, turned inward.

Sensing the inside

- Sherrington (1906) named the interoceptive senses. Interoception now means sensing the physiological state of the whole body: heartbeat, breathing, gut stretch, temperature, blood chemistry, inflammation, itch, pain.
- Three routes (Figure 7.2): the vagus nerve, about four in five of whose fibers carry signals toward the brain, to the nucleus of the solitary tract; thin spinal afferents to the dorsal horn, many to lamina I; and the circumventricular organs, where the blood-brain barrier is leaky and the brain samples the blood directly. All reach brainstem relays (the parabrachial nucleus), the hypothalamus and amygdala, and through the thalamus the posterior insula. Craig: the posterior insula holds a primary map of the body's state, re-represented and integrated toward the anterior insula.
- Measuring interoception: Schandry's heartbeat counting task. Critchley and colleagues (2004): right anterior insula activity and gray matter predict accuracy. Garfinkel and colleagues (2015): accuracy, believed ability, and how well confidence tracks accuracy need not agree.

- Sickness behavior (Dantzer 2008): cytokines from immune cells reach the brain (through the vagus and the circumventricular organs) and produce fatigue, loss of appetite, loss of pleasure, and withdrawal. Hart (1988): a strategy, not a weakness. Body signals reset priorities.
- Interoception is inference. Placebo analgesia after wisdom-tooth extraction, in the patients who responded to the placebo, is reversed by naloxone (Levine, Gordon, and Fields 1978): expectation released the brain's own opioids, which turned down the pain signal on its way to the brain. Seth, Barrett, and Simmons extend predictive coding inward: the insula compares predicted with arriving body signals. Kleckner and colleagues (2017) traced a network centered on cingulate cortex and insula that both regulates the viscera and receives their signals.

What an emotion is

- James (1884): we perceive the bear, the body reacts, and the feeling is our perception of the body's reaction. "We feel sorry because we cry." Lange made a similar proposal in 1885.
- Cannon (1927), five objections: cutting viscera from brain does not abolish emotional behavior; the same visceral changes occur in different emotions and in fever and exercise; viscera are poorly innervated; visceral changes are slow; adrenaline injections produce "as if" fear. (Adrenaline, which American physiologists call epinephrine, is a close relative of Chapter 3's norepinephrine.)
- Most objections have been qualified. The most direct modern test: Hsueh and colleagues (2023) paced mouse hearts with light, raising heart rate without touching the brain; in risky contexts only, the mice became more anxious, through the posterior insula. Garfinkel and colleagues (2014): fearful faces shown during the heart's contraction are detected more readily and judged more intense.
- Human tests that remove or impose bodily signals give smaller, less consistent effects: spinal cord injury reduces felt emotion little in a well-controlled study (Chwalisz 1988); the forced-smile effect failed a 17-laboratory replication (Wagenmakers 2016), though a meta-analysis finds a small facial feedback effect (Coles 2019).
- Arousal is interpreted. Schachter and Singer (1962): uninformed volunteers given adrenaline took on a confederate's mood (later replications weaker). Dutton and Aron (1974): half the men who took the interviewer's number on the swaying Capilano bridge called her, against about one in eight on a solid bridge.
- Three positions:

1) basic emotions (Ekman): a few emotions, each an evolved program with its own bodily pattern and universal facial expression;
2) constructed emotion (Barrett): the brain predicts and regulates the body and categorizes that state in context using learned concepts. Siegel and colleagues (2018): autonomic patterns overlap too much to identify emotions reliably. Barrett and colleagues (2019): facial movements are weak, context-dependent signs of emotion;
3) survival circuits (LeDoux): circuits for defense, feeding, reproduction, and fluid balance are well studied in animals; the word fear should be kept for the conscious feeling.

- Shared ground: all three can accept Russell's circumplex (1980), with valence and arousal as axes (core affect), as a description (Figure 7.3); and all agree that emotional states carry an estimate of how things stand for the body that sets priorities.

Value depends on the body

- Cabanac (1971), alliesthesia: a sweet solution is pleasant to a fasting person and unpleasant after a glucose load. Pleasure tracks usefulness to the body now.
- Rolls and colleagues (1981): sensory-specific satiety; the food being eaten loses pleasantness faster than foods not yet eaten.
- Betley and colleagues (2015): AgRP hunger neurons carry a negative teaching signal; mice avoid flavors and places paired with their activation. Eating is reinforcing partly because it ends the state.
- The reward prediction error that a large population of dopamine neurons signals (Chapter 3) is computed against outcomes whose value depends on the state of the body: juice is a large reward to a thirsty monkey and a small one to a sated monkey.
- Berridge and Robinson (1998) proposed, from dopamine-depleted rats that still showed liking reactions, separating wanting (dopamine-dependent motivational pull) differs from liking (hedonic impact, which survives dopamine loss and depends on opioid and other systems).

The amygdala and learned threat

- Klüver and Bucy (1939): monkeys without temporal lobes became tame, fearless, and oral. Later work tied much of the loss of fear to the amygdala.
- Fear conditioning (Figure 7.4): tone plus shock. The tone reaches the lateral amygdala directly from the auditory thalamus (fast, coarse) and through auditory cortex (slow, detailed); the shock reaches it through pain pathways; the tone synapses are strengthened by associative LTP (Chapter 5). The central nucleus drives the periaqueductal gray (freezing), the hypothalamus (blood pressure, stress hormones), and brainstem neuromodulatory nuclei.
- Romanski and LeDoux (1992): either route alone suffices for simple conditioning. Pessoa and Adolphs (2010): in primates the "low road" is not shown to be faster in practice or capable of fine discrimination. Méndez-Bértolo and colleagues (2016): human amygdala responses to fearful faces at about 74 ms, driven by coarse features. Unsettled.
- Not a fear center: Paton and colleagues (2006) found amygdala neurons for positive and negative value that changed with reversal.
- S.M. (Urbach-Wiethe disease, both amygdalae destroyed): almost no fear of snakes, spiders, a haunted house, or horror films (Feinstein 2011). But inhaling 35 percent carbon dioxide gave her and two other amygdala-damaged patients panic attacks, against three of twelve healthy volunteers (Feinstein 2013). Fear of threats from outside the body needs the amygdala; fear from inside the body, detected by chemical sensors outside the amygdala (probably in the brainstem), does not.
- Brief, cued fear depends on the central nucleus; sustained anxiety about uncertain or distant threats depends more on the bed nucleus of the stria terminalis (Davis 2010).

Unlearning fear

- Extinction is new learning, not forgetting. Bouton: extinguished fear returns with a change of context (renewal), with time (spontaneous recovery), or after an unsignaled shock (reinstatement).
- Milad and Quirk (2002): infralimbic neurons fire to the tone the day after extinction in proportion to how well it is remembered; stimulating them reduces freezing. In humans, the ventromedial prefrontal cortex (Phelps 2004).
- Exposure therapy is extinction in the clinic, with the same weakness: the fear can come back.
- The prefrontal cortex mostly inhibits the amygdala's output rather than erasing its memory (Chapter 5: fear is easily learned and rarely unlearned, only suppressed).
- Reappraisal (Ochsner 2002): reinterpreting a situation reduces negative feeling, raises prefrontal and lowers amygdala activity. Reappraisal is extinction made voluntary.

The chemistry of stress

- Two arms (Figure 7.5). Fast: sympathetic nerves and adrenaline from the adrenal medulla, in seconds. Slow: the hypothalamic-pituitary-adrenal (HPA) axis, CRH to ACTH to cortisol, peaking tens of minutes later; cortisol feeds back on hypothalamus, pituitary, and hippocampus to shut itself down.
- Selye (1936): the same triad (enlarged adrenals, shrunken thymus, ulcers) after many different insults. Sapolsky and colleagues (2000): cortisol's actions (permissive, suppressive, stimulatory, preparative, depending on timing) include restraining the stress response itself.
- Dum, Levinthal, and Strick (2019): neurons controlling the adrenal medulla trace back to motor cortex (trunk and posture) and cingulate regions involved in cognition and affect. Thought, movement, and stress are a few synapses apart.
- The settings are shaped by experience, starting early: rat pups licked and groomed more in their first days grow up with more hippocampal receptors for cortisol, stronger feedback, and smaller stress responses (Liu 1997). The response in the moment depends on company: holding a partner's hand reduced threat-related brain activity (Coan 2006).
- Inverted U (Figure 7.6). Yerkes and Dodson (1908): moderate shock best for a hard discrimination in mice (they varied shock and difficulty, not arousal; the arousal reading came decades later). Arnsten (2009): moderate norepinephrine and dopamine strengthen prefrontal working memory; high levels under stress weaken it and strengthen amygdala and habit circuits. Cortisol has its own inverted U through two receptors, one occupied at rest and one engaged under stress; memory is best when the first is filled and the second moderately engaged (de Kloet 2005).
- Allostatic load (McEwen): chronic activation is costly. Three weeks of restraint stress shrinks CA3 dendrites in the rat hippocampus (Magariños and McEwen 1995); chronic stress grows dendrites in the basolateral amygdala (Vyas 2002). Much of the hippocampal change reverses when stress ends; the amygdala growth and anxiety persisted three weeks into recovery (Vyas 2004).

Emotion marks what to remember

- Gold and van Buskirk (1975): adrenaline given after training improves memory, at moderate doses only. It changes what is kept, not what was experienced.
- McGaugh: adrenaline does not readily enter the brain; its effect is relayed, at least in part, by the vagus to the nucleus of the solitary tract (an interoceptive pathway), which releases norepinephrine in the basolateral amygdala, which modulates consolidation in hippocampus, striatum, and cortex.
- Cahill and colleagues (1994): a story with an upsetting middle is remembered better in the middle a week later, unless the listener took propranolol (a beta-adrenergic blocker) an hour before; memory for the neutral parts was unaffected. Since the drug was present during the story, this study does not separate encoding from consolidation; the animal work does. A patient with amygdala damage showed no enhancement (Cahill 1995).
- Arousal acts much like the third factor of Chapter 3's three-factor learning rule: it helps decide how strongly the traces are kept.
- Flashbulb memories (Brown and Kulik 1977) are confident, not accurate. Neisser and Harsch (Challenger, 1986); Talarico and Rubin (September 11, 2001): flashbulb memories lose consistency as fast as ordinary ones; what differs is belief in them.

Feeling and choosing

- Phineas Gage (1848): Harlow (1868) reported "a child in his intellectual capacity" with "the animal passions of a strong man", "no longer Gage". Earlier reports described a good recovery, and much of the legend is embellishment (Macmillan 2000). Skull reconstructions (H. Damasio 1994; Van Horn 2012) place the damage in ventromedial prefrontal cortex and underlying fiber tracts, more on the left.
- Elliot (A. Damasio 1994), a cleaner case than Gage: normal intelligence and reasoning, disastrous decisions, little feeling.
- Somatic marker hypothesis: anticipated bodily states bias choice and prune options.
- Iowa gambling task (Figure 7.7): decks A and B pay $100 per card but lose $250 per ten cards; C and D pay $50 and gain $250. Healthy players shift to C and D; ventromedial patients do not (Bechara 1994). Healthy players show anticipatory skin conductance responses before choosing bad decks, before they can say which are bad; patients do not, and some choose badly even when they know (Bechara 1997).
- The interpretation has not held up well: Maia and McClelland (2004) found players know which decks are good about as soon as they choose them; patients with pure autonomic failure play normally (Heims 2004); the patients' deficit is largely reversal learning and shrinks when the early large rewards are removed (Fellows and Farah 2005); Dunn and colleagues (2006) concluded the hypothesis needs more support to remain tenable.
- What survives: ventromedial damage leaves reasoning largely intact and choice impaired, and the region links value with the body. Whether bodily feedback is needed is doubtful; that value signals are tied to bodily regulation is not. Chapter 8 takes up the decision itself.

When regulation goes wrong

Working hypotheses, largely correlational (Khalsa 2018):
- Panic as a false suffocation alarm from an oversensitive suffocation monitor (Klein 1993); S.M.'s carbon dioxide panic fits.
- Anxiety as inflated interoceptive prediction error in the insula (Paulus and Stein 2006).
- Depression as a disorder of allostasis, a body predicted to need conserving (Barrett 2016); overlap with inflammation-induced sickness behavior.
- Alexithymia linked with poor interoception (Brewer 2016).

Treatments with good evidence (exposure therapy, reappraisal in cognitive behavioral therapy) can be read as ways of correcting the estimate. Propranolol is often prescribed for stage fright, but a systematic review found the evidence insufficient for routine use in any anxiety disorder (Steenen 2016).

Implications for artificial intelligence

1) Value comes from outside. RL rewards are specified by designers; an animal's are generated by its body. Homeostatic reinforcement learning (Keramati and Gutkin 2014): reward is the reduction in distance from set point. Singh and colleagues (2010): an internal reward that differs from the external goal can learn better.
2) A fixed external reward can be gamed (reward hacking, Amodei 2016). Animals are not immune (addiction), but values anchored to a body the brain cannot simply redefine are constrained.
3) Emotions are global modes. Simon (1967): an interrupt mechanism for a serial goal-pursuer. Doya (2002): neuromodulators set metaparameters (dopamine: prediction error; serotonin: time horizon; norepinephrine: explore versus exploit; acetylcholine: learning rate). Artificial agents mostly fix such parameters by hand.
4) Importance should decide what is learned. Prioritized experience replay (Schaul 2016) replays large-error transitions, but artificial agents rarely mark an experience as significant to the agent rather than merely surprising.
5) Reading emotion is not having it. Affective computing (named by Picard in the mid-1990s) often reads faces, and faces are weak, context-dependent signals (Barrett 2019).

Practical summary

- The brain regulates a body, and emotion is how that job reaches every other.
- Regulation is mostly predictive (allostasis): prepare before the disturbance arrives.
- Interoception runs through the vagus and lamina I to the insula and, like every sense, is inference.
- Bodily feedback changes feelings and perception (paced mouse hearts; fear faces at systole), but arousal is also interpreted in context.
- Basic, constructed, and survival-circuit views disagree about categories; valence and arousal are common ground.
- Value depends on bodily state (alliesthesia, sensory-specific satiety, hunger neurons).
- The amygdala learns the value of cues; extinction is new, prefrontal, context-dependent learning.

- Stress has a fast and a slow arm; its effects on the prefrontal cortex follow an inverted U; chronic stress reshapes hippocampus and amygdala in opposite directions.
- Arousal, through norepinephrine in the amygdala, is a third factor for consolidation; it strengthens conviction more than accuracy.
- Ventromedial prefrontal damage spares reasoning and impairs choice; the somatic marker account of why has not held up well.

Labs

Lab Notes: how to succeed in these labs

These labs turn the chapter's claims into small calculations. Each isolates one idea: predicting a disturbance versus correcting it, value that depends on the state of the body, measuring interoception, extinction as new learning, feedback in the stress axis, arousal as a gate on memory, and learning which options are good from their outcomes. The arithmetic is light; the point is to watch the rule behave and then ask what the biology gains or loses.

General rules:
1) Write every intermediate value, rounded to two decimal places unless told otherwise.
2) Draw the loop, circuit, or curve before computing.
3) End with an interpretation: what would change if a parameter changed, and what does the organism gain or lose?

What to hand in for every lab

- A table of values.
- A diagram or sketch where requested (a photo of a hand drawing is fine).
- A short interpretation (6-12 sentences).

Lab 7.1 - Correcting Versus Predicting (homeostasis and allostasis)

Purpose

To reproduce Figure 7.1 by hand: why a controller that acts on a predicted disturbance keeps a variable closer to its set point than one that waits for the error, and what happens when the prediction is wrong.

Given model

Blood glucose g (in mM) has set point 5. Each step, food adds d, and the controller removes u:
g_next = g + d − u
Reactive controller: u = k (g − 5), with k = 0.5, using the current g (it sees the error only once it exists).
Predictive controller: u = k (g − 5) + c d_cue, with c = 0.8, where d_cue is the amount of food the cue predicts for that step.
Meal: d = 0, 0, 2, 2, 1, then 0 for the remaining steps. Start with g = 5. Run ten steps.

Procedure

1) Reactive controller: tabulate g after each step. (Check: 5.00, 5.00, 7.00, 8.00, 7.50, 6.25, 5.62, 5.31, 5.16, 5.08; peak 8.00.)
2) Predictive controller, with the cue predicting the meal exactly (d_cue = d). (Check: 5.00, 5.00, 5.40, 5.60, 5.50, 5.25, 5.12, 5.06, 5.03, 5.02; peak 5.60.) By what factor is the peak excursion above the set point reduced?
3) Wrong prediction: the cue occurs but no food arrives (d = 0 throughout, d_cue as before). (Check: g falls to 3.40, then 2.60, minimum 2.60, before recovering.) Which experiment in the chapter is the mirror image of this case, with a drug instead of food and an unpredicted rather than a mispredicted dose?
4) Try to rescue the reactive controller by raising its gain to k = 1.2 with no cue. (Check: 5.00, 5.00, 7.00, 6.60, 5.68, 4.86, 5.03, …; peak 7.00.) Why can a larger gain not remove the first overshoot? What new problem does a high gain introduce (look at step 6)?
5) Write 6-10 sentences: what must the predictive controller know that the reactive one does not, where in the brain does that knowledge come from (relate it to the AgRP neurons of Chen and colleagues), and why does the chapter call allostasis “the same strategy, applied inward” as Chapter 6’s perception?

Lab 7.2 - Value Depends on the Body (alliesthesia by formula)

Purpose

To compute, from one simple rule, why the same food is a large reward, a small one, or a punishment depending on the state of the body (Cabanac’s sweet solution; Keramati and Gutkin’s homeostatic reinforcement learning).

Given model

An internal variable H (energy reserve) has set point 0. Drive is the squared distance from the set point: $D(H) = H^2$. An outcome that changes H by K has reward

$$\text{reward} = D(H) - D(H + K)$$

that is, reward is the reduction in drive it produces. A portion of food has K = +2.

Procedure

1) Compute the reward of the food at H = −6 (very hungry), −2, 0 (at set point), and +2 (overfed). (Check: 20, 4, −4, −12.)
2) Plot reward against H. Where does the food become a punishment? Relate this to Cabanac’s result and to sensory-specific satiety.
3) A hungry animal (H = −6) eats four portions in a row. Compute H and the reward of each portion. (Check: rewards 20, 12, 4, −4.) Explain in two sentences why the animal should stop after the third portion, and what tells it so.
4) Add a second variable, water W, with set point 0 and drive W^2. The animal is at H = −2, W = −6. Compare the reward of food (K = +2 on H) with water (+2 on W). Which does the animal value more, and by how much? (Check: food 4, water 20.)

5) A standard reinforcement learning agent is given a fixed reward of +5 for every portion of food. Describe in four sentences what it would do in step 3 and why that resembles reward hacking or addiction.
6) Write 6-10 sentences: which neurons in the chapter behave like the drive D (look at Betley and colleagues), why the dopamine reward prediction error of Chapter 3 needs the state of the body as an input, and what an artificial agent would need in order to have values of this kind.

Lab 7.3 - How Well Do You Feel Your Heart? (measuring interoception)

Purpose

To compute a heartbeat counting score, see why accuracy and confidence are different measures, and think about what the task really measures.

Given model

In the heartbeat counting task, a participant silently counts heartbeats over intervals of different length, without touching the pulse, while the true count is recorded. For each interval:
score = 1 − |actual − counted| / actual
and the participant's accuracy is the mean score over intervals (1 is perfect). After each interval the participant also rates confidence from 0 to 10.

Participant P: actual 30, 42, 55; counted 28, 40, 51; confidence 4, 5, 4.
Participant Q: actual 27, 38, 49; counted 15, 20, 24; confidence 8, 8, 9.

Procedure

1) Compute each interval score and the mean for both participants. (Check: P 0.933, 0.952, 0.927, mean 0.938; Q 0.556, 0.526, 0.490, mean 0.524.)
2) Garfinkel and colleagues distinguished accuracy (how well you count), sensibility (how good you believe you are), and awareness (how well confidence tracks accuracy). Describe P and Q on all three.
3) Q's counts are roughly half the true counts at every interval. Suggest two different explanations, one about the heart and one about strategy. What could a participant who simply knows that resting heart rate is about 70 beats per minute do to score well without feeling anything?
4) Optional, on yourself: sit quietly and count your heartbeats for 25, 35, and 45 seconds, without touching your pulse, while a partner takes your pulse at the wrist. Record your confidence. Compute your score. Do not share it with anyone who has not asked.
5) Write 6-10 sentences: why would the right anterior insula be expected to predict accuracy (Critchley and colleagues), and what would you need to show before claiming that better heartbeat perception causes better decisions in some job, rather than merely accompanying them?

Lab 7.4 - Extinction Is New Learning (a two-memory fear model)

Purpose

To show with a simple learning rule why extinguished fear returns in a new context (renewal) and with time (spontaneous recovery), and why a model in which extinction erases the original memory cannot explain this.

Given model

Acquisition: on each tone-shock trial the tone's association V is updated by the delta rule V ← V + a (1 − V), with a = 0.3, starting at V = 0. Fear to the tone equals V.
Extinction, in context B: the tone is presented alone. A separate inhibitory association I_B, tied to context B, is learned: fear in B = V − I_B, and after each trial I_B ← I_B + a × (fear in B). V itself is not changed.
In any context other than B, the inhibition is absent and fear = V.

Procedure

1) Run five acquisition trials. (Check: V = 0.30, 0.51, 0.66, 0.76, 0.83.)
2) Run eight extinction trials in context B. Tabulate fear after each trial. (Check: 0.58, 0.41, 0.29, 0.20, 0.14, 0.10, 0.07, 0.05; I_B = 0.78.)
3) Renewal: test the tone back in the training context A. (Check: fear = 0.83.) What would Bouton predict, and why does the model agree?
4) Spontaneous recovery: suppose that with the passage of a month the inhibitory memory weakens to half its strength. What is fear in context B? (Check: 0.44.)
5) Erasure model: instead of learning I_B, update V itself toward 0 during extinction, V ← V + a (0 − V). Show that it produces exactly the same extinction curve as step 2 (check: 0.58, 0.41, … 0.05), and then show that it predicts no renewal and no spontaneous recovery. What does this tell you about using an extinction curve alone to decide between theories?
6) Where in the brain is I_B (Milad and Quirk; Phelps and colleagues), and where is V? Draw the circuit of Figure 7.4 and mark both.
7) Write 6-10 sentences: why is it adaptive for the brain to keep the original fear memory, what does this imply for exposure therapy (where should it be done, and how often), and how does reappraisal resemble extinction?

Lab 7.5 - The Stress Axis and Its Brake (negative feedback)

Purpose

To see how cortisol's feedback on the brain limits the stress response and sets its baseline, and what happens when the feedback weakens.

Given model

Cortisol C is cleared at 30 percent per step and produced in response to a hypothalamic drive h:
h = max(s + 1 − f C, 0)
C_next = 0.7 C + h

where s is the stressor (10 on steps 1 to 3, 0 afterward), 1 is a constant basal drive, and f is the strength of feedback from cortisol onto hypothalamus, pituitary, and hippocampus. Before the stressor, C sits at its baseline, 1 / (0.3 + f).

Procedure

1) Compute the baseline for f = 0.5 and for f = 0.1. (Check: 1.25 and 2.50.)
2) With f = 0.5, start at baseline and run sixteen steps. (Check: 11.25, 13.25, 13.65, 9.55, 6.69, 4.68, 3.28, 2.29, 1.61, 1.32, 1.26, 1.25, …; peak 13.65.)
3) With f = 0.1, repeat. (Check: 12.50, 18.50, 22.10, 15.47, 10.83, 7.58, 5.55, 4.33, 3.60, 3.16, 2.90, 2.74, …; peak 22.10.)
4) For each, at which step does C first come within 0.5 of its own baseline after the stressor? (Check: step 9 for f = 0.5, step 11 for f = 0.1.) State three differences between the two runs.
5) Chronic stress shrinks hippocampal dendrites (Magariños and McEwen), and the hippocampus is one of the feedback sites. Use the model to explain why chronic stress might make the next stress response larger, longer, and higher at rest. Name this vicious circle in the chapter's terms.
6) The sympathetic arm acts in seconds and the HPA arm in tens of minutes. Draw both on a single time axis for one stressful event, using Figure 7.5.
7) Write 6-10 sentences: why does the body need a brake on a response that is helpful in the short term, why is cortisol called preparative and permissive, and why does the inverted U of Figure 7.6 mean that the right amount of stress hormone depends on the task?

Lab 7.6 - Arousal Decides What Is Kept (a three-factor consolidation rule)

Purpose

To model the Cahill propranolol experiment as a three-factor rule in which arousal, acting through beta-adrenergic receptors, gates how much of an event's trace is consolidated.

Given model

Each event leaves a trace of strength 1. The fraction consolidated is
kept = 0.2 + 0.6 × arousal × b
where arousal is between 0 and 1, and b is beta-adrenergic receptor availability: 1 with placebo, 0 with propranolol. The neutral parts of the story have arousal 0.1; the upsetting middle has arousal 0.9.

Procedure

1) Compute kept for the neutral and emotional parts, with placebo and with propranolol. (Check: placebo 0.26 and 0.74; propranolol 0.20 and 0.20.) In the real study the drug was taken an hour before the story; the model treats it as acting only on consolidation. What experiment would justify that?
2) Which features of Cahill and colleagues' result does the model reproduce? Which term in the rule corresponds to the amygdala, and which to the hippocampus and cortex where the memory is stored?

3) Gold and van Buskirk found that adrenaline after training helped at moderate doses and not at high ones. Modify the rule so that kept follows an inverted U in arousal. Write your rule and compute it at arousal 0.1, 0.5, and 0.9.
4) A patient whose amygdalae are destroyed (Cahill 1995) has b = 1 but no arousal signal reaching the storage sites. Which term do you set to 0, and what does the patient remember?
5) The model says nothing about accuracy. Using the flashbulb studies, add a second quantity, confidence, that grows with arousal even while consistency decays over weeks at the same rate for both kinds of memory. Sketch both curves for an emotional and a neutral event.
6) Write 6-10 sentences: name the three factors of Chapter 3's rule in this setting, explain why a brain should keep emotionally arousing events preferentially, and explain why preferential keeping does not guarantee accuracy.

Lab 7.7 - Four Decks (the Iowa gambling task by hand)

Purpose

To compute why decks A and B are bad and C and D good, to watch a simple learner discover it from outcomes, and to see what a learner that ignores delayed losses would do.

Given model

Deck A pays 100 on every card, with losses on cards 3, 5, 7, 9, and 10 of 150, 300, 200, 250, and 350. Deck C pays 50 on every card, with a loss of 50 on cards 3, 5, 7, 9, and 10. (Decks B and D have the same totals as A and C with fewer, larger losses.)
A learner keeps a value for each deck and after each card updates it with $V \leftarrow V + 0.2\,(r - V)$, where r is the card's net result (win minus loss), starting at V = 0.

Procedure

1) Compute the net result over ten cards and the expected value per card for decks A and C. (Check: A −250 and −25; C +250 and +25.)
2) Draw ten cards from deck A and update its value after each. (Check: 20.0, 36.0, 18.8, 35.0, −12.0, 10.4, −11.7, 10.7, −21.5, −67.2.) Repeat for deck C. (Check: 10.0, 18.0, 14.4, 21.5, 17.2, 23.8, 19.0, 25.2, 20.2, 16.1.)
3) After how many cards is deck A's value first below deck C's? Is it below at every later card? What does this say about how long a learner needs before the bad deck is reliably recognized?
4) A myopic learner updates only on wins and ignores losses. Which deck does it prefer, and by how much after ten cards from each? (Check: A 89.3, C 44.6.) Relate this to Bechara's description of patients as "insensitive to future consequences."
5) Bechara and colleagues read healthy players' anticipatory skin conductance responses as a bodily signal of the learned value. Maia and McClelland argue that players also know the values consciously. In this model, V is just a number. What experiment would tell whether V is felt, known, or both?
6) Fellows and Farah found that the patients' deficit shrank when the bad decks no longer paid well at the start. Rerun step 2 for deck A with its cards in a different order: the five losing cards first (cards 1 to 5 are the losses, cards 6 to 10 pure wins). When does A's value first fall below C's

now? (Check: from the first card; A reaches −111.4 after card 5, but climbs back to 30.7 by card 10, above C's 16.1.) What does the late reversal say about a learner with a fixed learning rate of 0.2? Explain why a learner that is slow to reverse would suffer most from the original order.

7) Write 6-10 sentences: what the Iowa gambling task shows about the ventromedial prefrontal cortex, what the pure autonomic failure patients imply about the role of the body, and how Chapter 8 might frame the same task as a decision under uncertainty.

Lab 7.8 - From Body to Choice (concept map)

Purpose

To integrate the chapter on one page: how the state of the body becomes a feeling, a value, a prepared body, a remembered event, and a choice.

Procedure

1) Draw a body with heart, lungs, gut, and immune cells. Draw the two interoceptive routes (vagus to the nucleus of the solitary tract; lamina I) to the brainstem, hypothalamus, amygdala, thalamus, and posterior and anterior insula (Figure 7.2).
2) Draw the hypothalamus as a controller with a set point, and add a predictive input from cortex labeled "allostasis: act before the error."
3) Draw the downward arms: sympathetic nerves and adrenaline, and the HPA axis with cortisol feedback (Figure 7.5).
4) Add the amygdala circuit for learned threat (Figure 7.4), with the ventromedial prefrontal cortex inhibiting it after extinction.
5) Add Chapter 3's neuromodulators: norepinephrine from the locus coeruleus driving consolidation through the amygdala, and dopamine carrying a reward prediction error whose value depends on bodily state.
6) Add the valence and arousal axes as a small inset (Figure 7.3), with an arrow from the insula labeled "core affect."
7) Draw the ventromedial prefrontal cortex receiving interoceptive and value signals and sending a bias to choice, with a note on the Iowa gambling task and the evidence against the somatic marker reading of it.
8) Write a 10-14 sentence caption that follows one event, such as a near miss in traffic, from the first sight of the car to the memory of it a week later, naming the mechanism at each step and one place where a disorder of regulation would change the outcome.